

- 1 Fig. 1.1 is a transmission electron micrograph of cells from duckweed, *Spirodela oligorrhiza*.

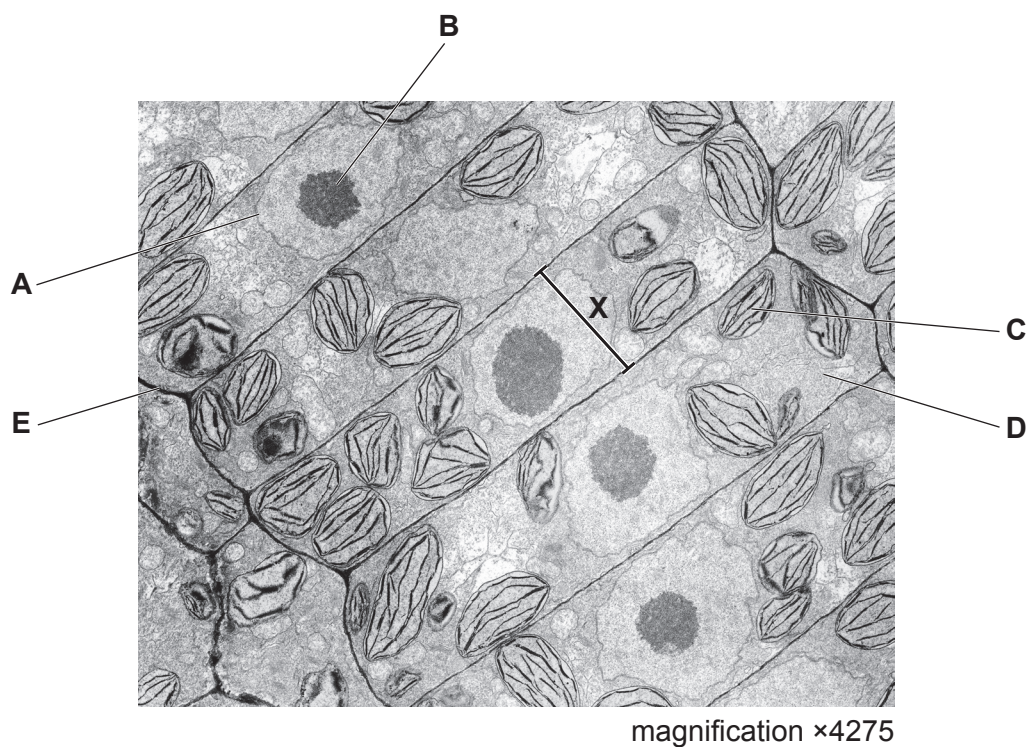


Fig. 1.1

- (a) Calculate the actual width of the cell labelled X.

Write down the formula you will use to make your calculation.

Show your working and give your answer in micrometres to one decimal place.

formula

..... μm [3]

- (b) (i) Table 1.1 lists some biological molecules found in plant cells.

Complete Table 1.1 by choosing **one** letter from Fig. 1.1 that indicates a cell structure where each biological molecule is found.

Table 1.1

biological molecule	letter from Fig. 1.1
DNA	
cellulose	
phospholipid	
histone proteins	

[4]

- (ii) State the name of a cell structure, **visible in Fig. 1.1**, where ATP is synthesised.

..... [1]

- (iii) Name a cell structure that produces mRNA.

..... [1]

- (c) Describe the evidence from Fig. 1.1 that shows that the image is a transmission electron micrograph.

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..... [2]

[Total: 11]